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UNNI C02 SMARTIFICATION — REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32-C6 + Auxiliary

Simulation Harness

File: 01_usb_power.kicad_sch

File: 02_esp_aux.kicad_sch

File: 03_simulation_harness.kicad_sch

Unni C02 Smartification

Sheet: /
File: unni-smartification-c6.kicad_sch

Title: Unni C02 Smartification — ESP32-C6 Rev A

Size: A3Date: 2026-08-18

KiCad E.D.A. 10.0.5Id: 1/4

A

B

C

D

E

F

A

B

C

D

E

F

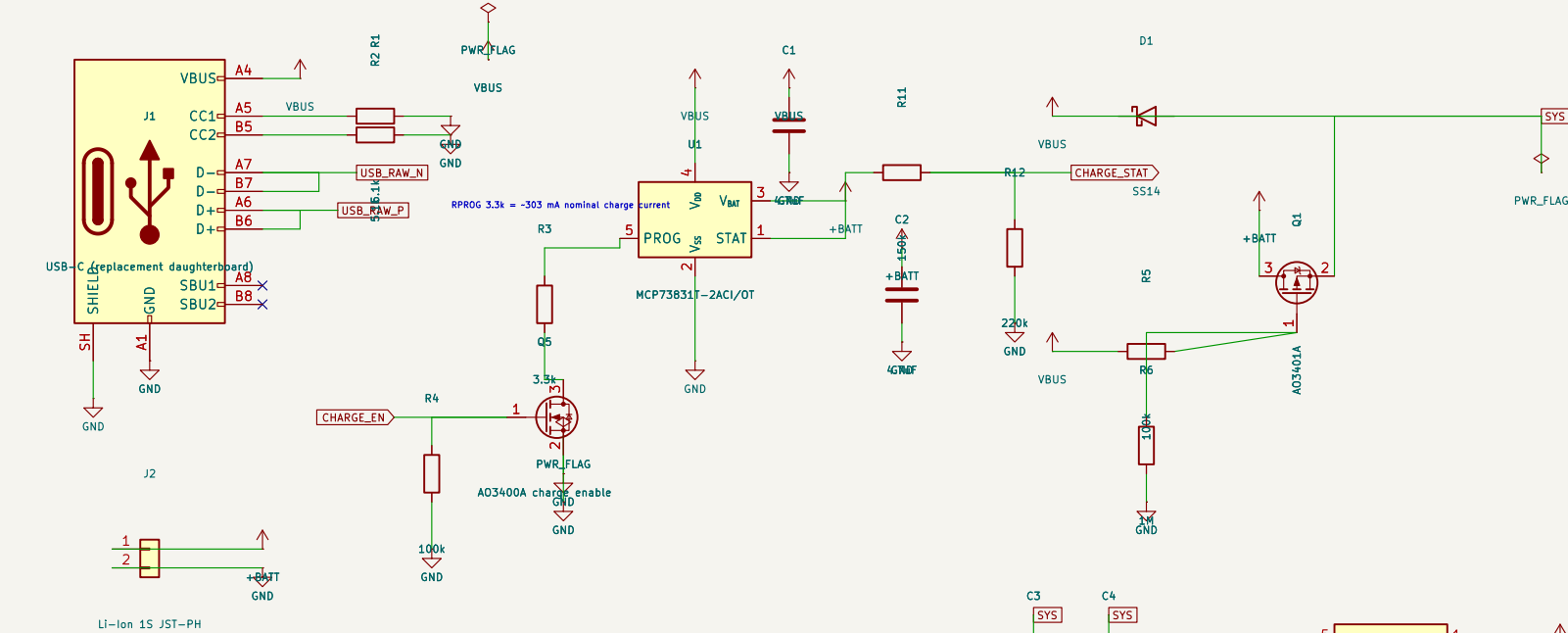
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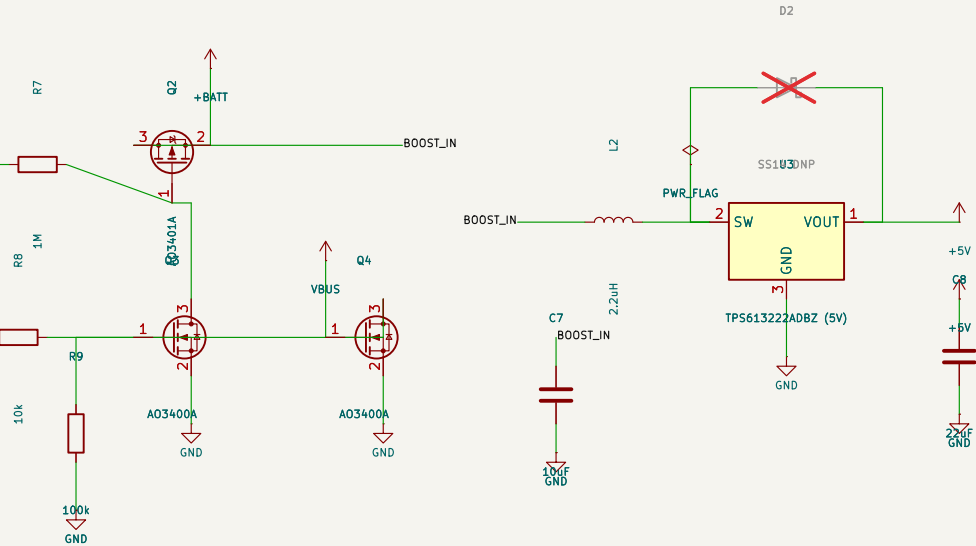
Unni C02 Smartification			
Sheet: / File: unni-smartification-c6.kicad_sch			
Title: Unni C02 Smartification — ESP32-C6 Rev A			
Size: A3	Date: 2026-08-18		Rev: A-v26-kicad-sim
KiCad E.D.A. 10.0.5	Id: 1/4		

USB + POWER

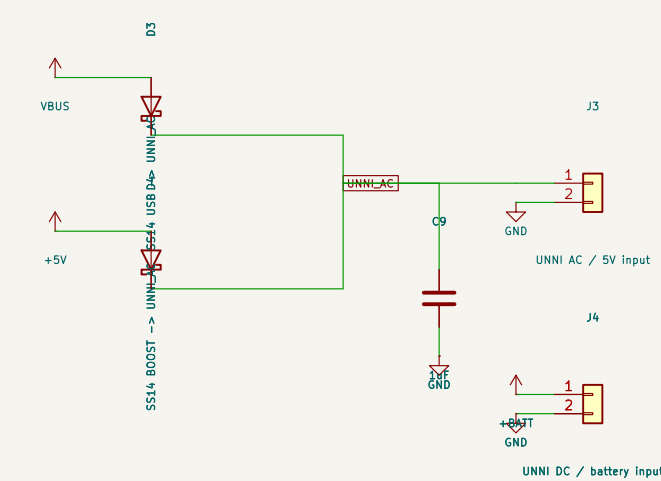
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



FORCED-AWAKE 5V



UNNI POWER OUTPUTS



Unni C02 Smartification

Sheet: /USB + Power/
File: 01_usb_power.kicad_sch

Title: USB + Power

Size: A3
Date: 2026-08-18

Rev: A-v26-kicad-sim
Id: 2/4

ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.

A

B

C

D

E

F

A

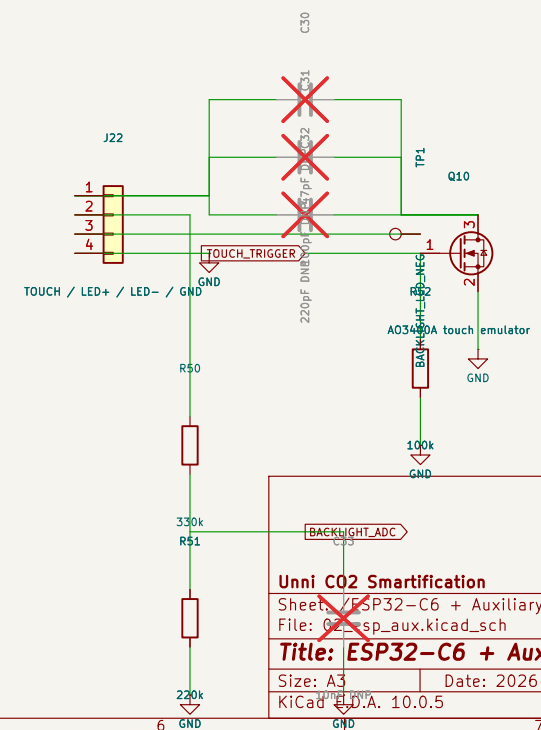
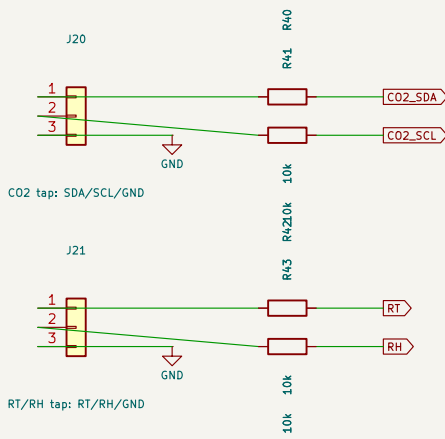
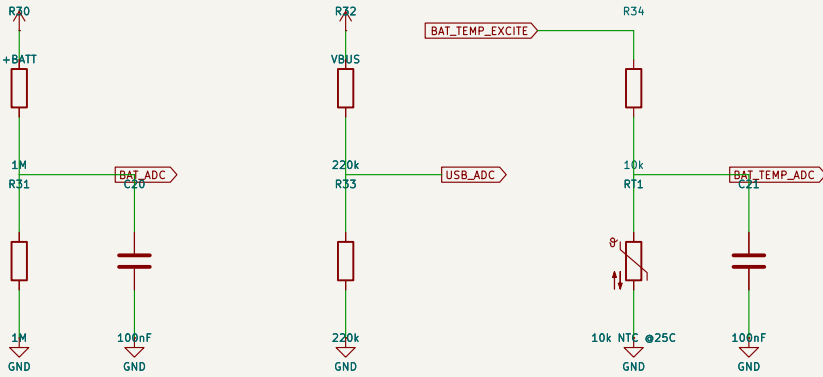
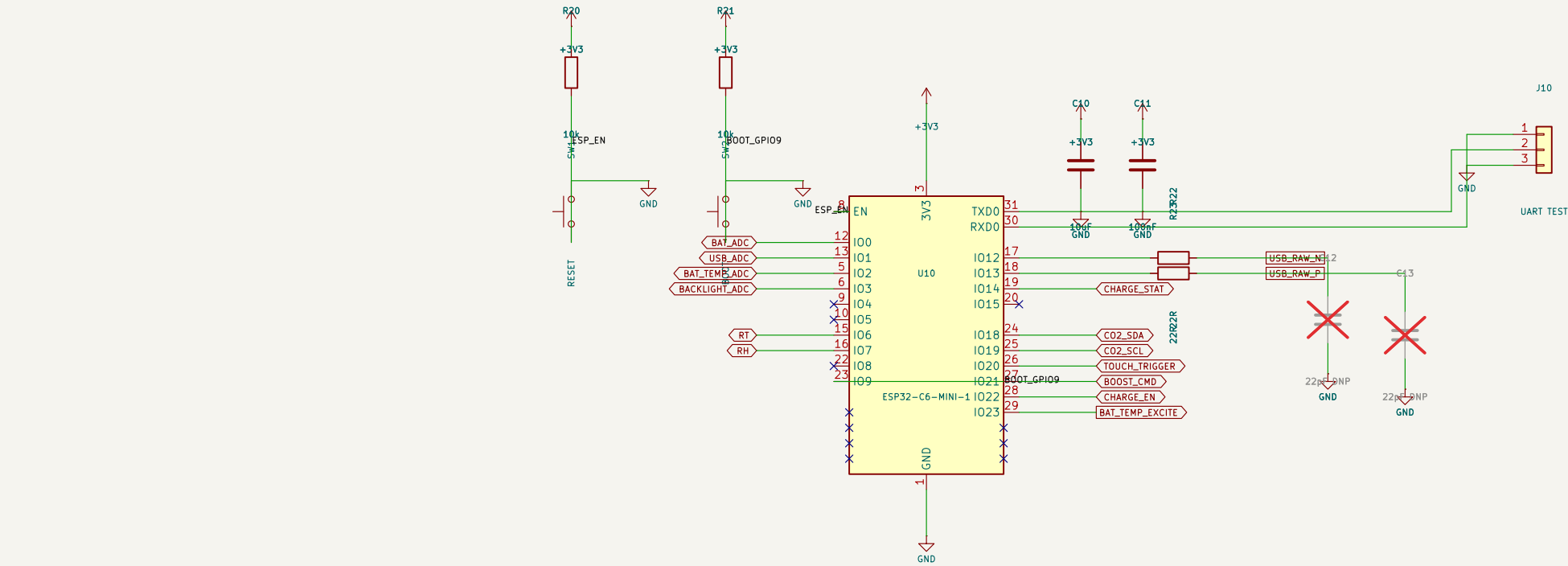
B

C

D

E

F



1	2	3	4	5	6	7	8
SIMULATION HARNESS – excluded from BOM/PCB							

Scenario: battery-only → forced awake → USB hot-plug while BOOST_CMD remains asserted → charge enable → USB removal. Load values are validation placeholders.

Unni C02 Smartification

Sheet: /Simulation Harness/
File: 03_simulation_harness.kicad_sch

Title: Simulation Harness

Size: A3	Date: 2026-08-19	Rev: A-v26-kicad-1
KiCad E.D.A. 10.0.5		Id: 4/4

Rev: A-v26-kicad-s
Id: 4/4